

4J Time - Extra Questions

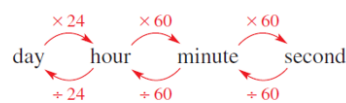
Learning Objective	Solve problems of duration involving 12- and 24-hour cycles across multiple time zones
Today We Are	Solving problems involving time
Indicators of Progress	• I can convert to 24 hour time • I can calculate the time difference

Key Ideas

■ The standard unit of time is the **second** (s).

■ Units of time include:

- 1 **minute** (min) = 60 seconds (s)
- 1 **hour** (h) = 60 minutes (min)
- 1 **day** = 24 hours (h)
- 1 **week** = 7 days
- 1 **year** = 12 months



■ **24-hour time** shows the number of hours and minutes after midnight.

- 0330 is 3:30 a.m.
- 1530 is 3:30 p.m.

■ The 'degrees, minutes and seconds' button on a calculator can be used to convert a particular time into hours, minutes and seconds.

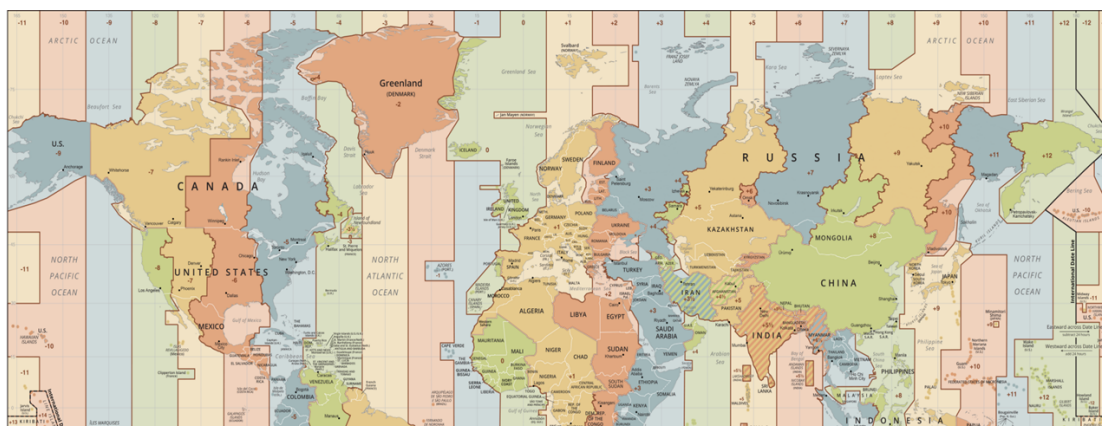
For example: 4.42 hours = 4°25' 12" meaning 4 hours, 25 minutes and 12 seconds

■ Earth is divided into 24 time zones (one for each hour).

- Twenty-four 15° lines of longitude divide Earth into its time zones. Time zones also depend on a country's borders and its proximity to other countries. (See the map on pages 296–7 for details.)
- Time is based on the time in a place called Greenwich, United Kingdom, and this is called Coordinated Universal Time (UTC) or Greenwich Mean Time (GMT).
- Places east of Greenwich are ahead in time.
- Places west of Greenwich are behind in time.

■ Australia has three time zones:

- Eastern Standard Time (EST), which is UTC plus 10 hours.
- Central Standard Time (CST), which is UTC plus 9.5 hours.
- Western Standard Time (WST), which is UTC plus 8 hours.





Example 1

Calculate the time

a) 3 hours after 09:25

$$9.25 + 3 \text{ hrs} = 12.25$$

b) $1 \frac{3}{4}$ hours before 17:00

$$17.00 - 1 \frac{3}{4} \text{ hours} = 15.15$$

i
45 minutes

c) 95 minutes after 07:40

$$07.40 + 60 \text{ minutes} = 08.40$$

60 35

$$8.40 + 35 \text{ minutes} = 09.15$$

Example 2

a) Liam's flight leaves at **6:50 am** and lands at **10:35 am** in the same time zone.
How long is his flight?

$$6.50 \rightarrow 7.00 = 10 \text{ minutes}$$

$$7.00 \rightarrow 10.35 = 3 \text{ hrs } 35 \text{ minutes}$$

$$\text{Total time} = 3 \text{ hrs } 45 \text{ minutes}$$

b) Ava flies from **8:45 am** to **3:30 pm**. 15.30
Calculate total flight time.

$$8.45 \rightarrow 9.00 = 15 \text{ minutes}$$

$$9.00 \rightarrow 15.30 = 6 \text{ hrs } 30 \text{ minutes}$$

Total time = 6 hrs 45 minutes

c) Ethan starts homework at 10:12 am and finishes at ~~2:35~~ pm. 14:35

10:12 $\rightarrow$ 10:35 = 23 minutes
10:35 $\rightarrow$ 14:35 = 4 hours

Total time = 4 hours 23 minutes

Example 3

Dean travels from Brisbane to New York.

Activity	Duration
Flight: Brisbane $\rightarrow$ Singapore	8 hours, 52 minutes
Stop over	14 hours, 40 minutes
Flight 2: Singapore $\rightarrow$ New York	16 hours, 24 minutes

What is the total travel time for Daniel in days, hours and minutes?

8 + 14 + 16 = 38 hours
 $\wedge$
 24 14

= 1 day 14 hours

52 + 40 + 24 = 116 minutes
 $\wedge$
 60 56

= 1 hr 56 minutes

Total time = 1 day 14 hours + 1 hr 56 minutes
= 1 day 15 hours 56 minutes

Time & Travel Questions

A) Calculate the time

1. 3 hours after 07:25
2. 1 $\frac{1}{2}$ hours before 16:00
3. 80 minutes after 07:35
4. 2 $\frac{1}{4}$ hours after 12:40

B) Flight duration questions

5. A flight leaves at **07:15 am** and arrives at **11:45 am** (same time zone).
 How long is the flight?
6. A plane departs at **13:20** and lands at **20:05**.
 Calculate the flight time.
7. A flight leaves at **22:10** and arrives at **05:40 (next day)**.
 How long is the flight?

C) Time duration (start → finish)

8. Alaya starts homework at **11:17 am** and finishes at **4:10 pm**.
 How long did she work?
9. Jacob studies from **15:45** to **19:25**.
 How long did he study?

D) Multi-stage travel (flight + stopover)

10. Sophie travels from Brisbane to Paris.

Activity	Duration
Flight: Brisbane → Shanghai	10 hours, 30 minutes
Stopover	2 hours, 14 minutes
Flight 2: Shanghai → Beijing	6 hours, 40 minutes

-  What is the total travel time in **days, hours and minutes**?

11. Daniel travels from Brisbane to New York.

Activity	Duration
Flight: Brisbane → Singapore	8 hours, 55 minutes
Stopover	14 hours, 40 minutes
Flight 2: Singapore → New York	16 hours, 25 minutes

👉 What is the total travel time in **days, hours and minutes**?

12. Mia travels from Brisbane to Rome.

Activity	Duration
Flight: Brisbane → Dubai	14 hours, 10 minutes
Stopover	5 hours, 35 minutes
Flight 2: Dubai → Rome	6 hours, 50 minutes
Stopover 2: Rome airport delay	3 hours, 25 minutes

👉 What is the total travel time in **days, hours and minutes**?